

YELABUGA, Russian Federation

WMO: 317330

Lat: 48.8034N

Lon: 135.8740E

Elev: 191

StdP: 14.59

Time Zone: 10.00 (E10)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-17.3	-26.1	1.3	-18.2	-23.3	1.5	-16.3	18.6	0.0	16.2	-0.8	6.3	160	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	84.2	72.1	81.7	70.8	79.5	69.5	74.6	80.9	72.9	78.7	71.4	77.0	5.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.7	121.9	77.8	71.0	115.3	76.3	69.4	109.0	74.8	38.4	81.5	36.9	79.0	35.4	77.0	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.3	13.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.6	-3.7	3.1	21.1	40.5	54.7	64.7	71.1	68.2	57.9	40.8	18.4	0.3
	DBStd	27.67	8.14	9.04	9.99	7.67	7.80	6.26	4.87	4.83	6.48	8.14	11.71	8.38
	HDD50	7024	1665	1312	895	303	44	0	0	0	11	304	948	1542
	HDD65	10786	2130	1732	1360	735	328	82	14	25	225	750	1398	2007
	CDD50	2131	0	0	0	18	189	441	653	563	248	19	0	0
	CDD65	416	0	0	0	0	7	73	201	123	12	0	0	0
	CDH74	2350	0	0	0	5	99	510	1141	549	46	0	0	0
	CDH80	410	0	0	0	0	10	120	219	59	2	0	0	0
	CDH80	410	0	0	0	0	10	120	219	59	2	0	0	0
(14) Wind	WSAvg	6.9	6.9	6.9	6.8	7.5	7.3	5.7	5.4	5.3	6.2	8.4	8.4	7.5
(15) Precipitation	PrecAvg	25.2	0.6	0.6	1.0	1.4	2.4	2.7	4.8	5.1	3.2	1.7	1.1	0.8
	PrecMax	35.9	1.4	1.7	2.4	3.7	5.6	6.7	9.4	9.9	7.3	3.8	2.4	1.9
	PrecMin	16.4	0.0	0.0	0.1	0.3	0.7	0.2	0.7	1.3	0.8	0.2	0.2	0.2
	PrecStd	4.8	0.4	0.4	0.7	0.7	1.1	1.6	2.3	2.1	1.5	0.9	0.7	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	33.6	49.7	72.5	81.0	87.7	88.1	84.2	78.9	70.7	47.6	26.2
		MCWB	17.7	27.6	39.8	53.9	62.1	70.8	74.7	73.9	68.2	58.2	40.0	23.5
	2%	DB	13.9	27.4	44.0	65.7	76.8	83.7	84.6	81.6	74.7	63.5	41.8	19.0
		MCWB	12.3	23.9	36.0	50.1	58.5	69.4	73.2	72.1	65.5	53.3	37.0	16.9
	5%	DB	10.9	22.2	39.2	59.2	72.6	80.1	82.3	79.4	71.4	58.4	37.3	15.3
		MCWB	9.3	19.4	32.6	46.0	56.5	67.7	72.3	70.4	63.6	48.7	32.6	13.8
	10%	DB	7.7	17.4	35.5	53.6	69.0	76.4	80.1	77.1	68.5	53.5	33.9	12.6
		MCWB	6.4	15.1	30.5	42.3	55.6	66.5	71.1	69.3	61.8	45.9	31.2	11.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	28.1	40.2	55.8	63.7	72.4	76.5	76.3	69.7	58.3	42.1	23.5
		MCDB	19.2	32.6	49.2	69.6	75.5	83.1	82.0	81.1	74.4	67.6	46.7	26.5
	2%	WB	12.4	24.0	36.2	50.3	61.1	70.5	75.3	74.1	67.4	53.9	37.3	17.1
		MCDB	14.0	27.3	42.9	64.0	73.1	80.6	82.2	79.0	72.6	62.4	42.0	18.7
	5%	WB	9.3	19.6	33.4	46.4	59.0	69.0	73.9	72.2	65.2	50.3	33.9	13.9
		MCDB	10.7	21.9	38.2	57.7	68.7	78.0	80.2	77.1	70.0	56.7	36.1	15.2
	10%	WB	6.5	15.3	30.9	43.5	56.7	67.4	72.4	70.5	62.8	46.7	31.5	11.3
		MCDB	7.8	17.4	35.1	51.8	66.8	75.1	78.1	75.0	67.3	52.9	33.9	12.6
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	15.2	15.2	16.8	17.4	15.7	14.2	13.4	16.1	14.9	12.1	11.2
		MCDBR	13.9	18.7	17.2	25.8	24.6	20.8	17.7	17.7	19.3	22.4	14.5	12.9
		MCWBR	12.6	15.5	11.5	14.0	11.6	9.2	8.5	8.6	10.0	13.4	10.8	11.5
		MCDBR	13.7	18.0	15.7	24.3	20.4	18.0	15.5	14.2	16.5	20.5	13.1	12.8
	5% WB	MCWBR	12.4	15.0	10.6	13.7	10.6	8.7	8.5	8.2	9.3	12.9	10.0	11.6
(40) Clear-Sky Solar Irradiance	taub		0.288	0.335	0.408	0.528	0.562	0.511	0.502	0.445	0.411	0.440	0.375	0.302
	taud		2.076	1.943	1.824	1.751	1.772	1.997	2.067	2.236	2.237	1.973	2.002	2.051
	Ebn at Noon		235	246	246	228	227	242	242	252	248	206	192	207
	Edh at Noon		28	41	55	66	67	54	50	40	36	40	30	25
(44) All-Sky Solar Radiation	RadAvg		575	965	1417	1514	1667	1784	1659	1452	1180	785	506	437
	RadStd		23	31	89	149	139	215	146	99	90	58	40	24

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page